Report #3: Cloud Resource Governance and Tenant Quota Enforcement

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Scenario

Following the successful remediation of storage functionality and implementation of tenant network isolation in Report #2, Southriver Software Solutions now operates a functional multi-tenant private cloud capable of supporting development and testing workloads. However, operational testing revealed an additional enterprise risk: unrestricted resource consumption across tenant projects.

Although DEV and QA environments are now isolated at the network layer, both projects retain unrestricted access to compute, storage, and networking resources. Without governance controls, a single tenant could unintentionally exhaust shared infrastructure capacity through excessive instance deployment, volume allocation, or floating IP usage.

Industry cloud providers mitigate this risk through quota enforcement mechanisms, which limit resource consumption per tenant while maintaining operational availability [1].

This engagement focuses on implementing project-level governance controls within OpenStack to prevent resource exhaustion, enforce fair usage, and validate denial behavior when limits are exceeded [1].

Prerequisites

- Functional OpenStack DevStack deployment
- Completed Reports #1 and #2 configuration
- Administrative OpenStack access
- Existing DEV and QA projects with deployed instances
- Familiarity with OpenStack CLI operations

Learning Objectives

By completing this lab, students will be able to:

- Understand risks associated with unrestricted tenant resource consumption

- Configure project-level quotas in OpenStack
- Enforce governance controls across compute, storage, and networking services
- Validate quota enforcement through controlled failure testing
- Demonstrate cloud resource protection aligned with enterprise practices

Background

Previous engagements established identity controls, restored persistent storage, and implemented tenant network segmentation to prevent lateral movement between environments. Report #2 confirmed that DEV and QA workloads operate within isolated Neutron networks protected by restrictive security groups.

While these controls protect confidentiality and access boundaries, they do not prevent denial-of-service conditions caused by excessive internal resource allocation. In shared cloud environments, uncontrolled provisioning may degrade performance or render infrastructure unavailable to other tenants [1].

OpenStack addresses this operational risk through quota management, allowing administrators to define consumption limits per project across compute, block storage, and networking services [1] [2] [3].

Business Scenario

Southriver Software Solutions expects multiple development teams to share cloud infrastructure. During internal testing, QA engineers unintentionally deployed numerous test instances, consuming excessive compute capacity and reducing availability for development workloads.

Southriver has requested the consulting team implement governance safeguards ensuring that no single tenant can monopolize shared infrastructure resources while preserving normal operational workflows.

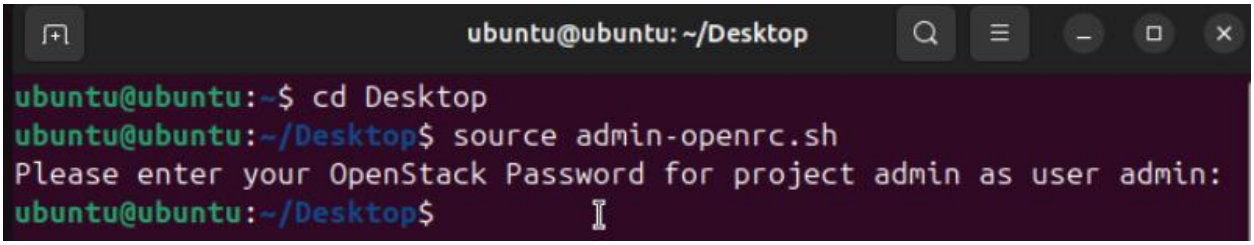
Section 1: Baseline Resource Consumption Assessment

Purpose of This Section

This section documents the current unrestricted resource allocation model prior to governance enforcement. Establishing a baseline demonstrates the risks associated with unlimited tenant provisioning.

1.1 Authenticate as Administrator

Step 1: Open a terminal and authenticate using administrative credentials. Sourcing the admin credentials scopes all subsequent CLI operations to the administrative context, which is required to view and modify project quotas.



You are now authenticated as the OpenStack administrator.

1.2 Review Existing Compute Quotas

Step 2: Display current quota limits for the DEV project. This captures the default quota configuration before any governance controls are applied, providing the before-state for comparison.

```
ubuntu@ubuntu:~/Desktop$ openstack quota show dev
+-----------------------+-------+
| Resource              | Limit |
+-----------------------+-------+
| cores                 |    20 |
| instances             |    10 |
| ram                   | 51200 |
| fixed_ips             |  None |
| networks              |   100 |
| volumes               |    10 |
| snapshots             |    10 |
| gigabytes             |  1000 |
| backups               |    10 |
| volumes_lvmdriver-1   |    -1 |
| gigabytes_lvmdriver-1 |    -1 |
| snapshots_lvmdriver-1 |    -1 |
| volumes___DEFAULT__   |    -1 |
| gigabytes___DEFAULT__ |    -1 |
| snapshots___DEFAULT__ |    -1 |
| groups                |    10 |
| check_limit           |  None |
```

Step 3: Repeat for QA. Comparing default quotas across both projects confirms that both are operating under the same permissive baseline.

```
ubuntu@ubuntu:~/Desktop$ openstack quota show qa
+-----------------------+-------+
| Resource              | Limit |
+-----------------------+-------+
| cores                 |    20 |
| instances             |    10 |
| ram                   | 51200 |
| fixed_ips             |  None |
| networks              |   100 |
| volumes               |    10 |
| snapshots             |    10 |
| gigabytes             |  1000 |
| backups               |    10 |
| volumes_lvmdriver-1   |    -1 |
| gigabytes_lvmdriver-1 |    -1 |
| snapshots_lvmdriver-1 |    -1 |
| volumes___DEFAULT__   |    -1 |
| gigabytes___DEFAULT__ |    -1 |
| snapshots___DEFAULT__ |    -1 |
| groups                |    10 |
| check_limit           |  None |
```

Observation

Projects typically inherit default quotas permitting large numbers of:

- Instances
- vCPUs
- RAM allocation
- Floating IPs
- Security group rules [1] [3]

Document that current limits exceed realistic organizational needs.

1.3 Demonstrate Unrestricted Instance Creation

Step 4: Switch to the DEV project. Scoping the session to DEV allows instance creation to be performed as a tenant user, demonstrating the risk from a tenant perspective rather than an admin one.

```
ubuntu@ubuntu:~/Desktop$ export OS_PROJECT_NAME=dev
ubuntu@ubuntu:~/Desktop$ unset OS_PROJECT_ID
ubuntu@ubuntu:~/Desktop$ openstack token issue
+------------+-------------------------------------------------------+
| Field      | Value                                                 |
+------------+-------------------------------------------------------+
| expires    | 2026-02-22T21:36:57+0000                              |
| id         | gAAAAABpm2jpDygXoLcIoQ5u-                             |
|            | bo18_nvzq8XELHosUwH3V1PPrQl7T50D3BcWuzD-              |
|            | 6mbiSwfnww7sJrB0ww6PYhsnsqLkJGCpB6AYfopp_FhK_Rif4MOoe |
|            | PaLkBU7lA4OvhhQhdf2C23-                               |
|            | AaQPCkUMEF5tNMH2M5gK4zz5HzxgHSiGJVCeaZHkAA            |
| project_id | 4f97f0eebec74d8b83b9296bc70dd0bb                      |
| user_id    | 2eee114ca75f4851850324b95989f70f                      |
+------------+-------------------------------------------------------+
```

Step 5: Create multiple test instances, repeating instance creation up to 10 times and changing the server name each time. This demonstrates that without quota controls, a tenant can provision resources without any system-enforced limit.

```
ubuntu@ubuntu:~/Desktop$ openstack server create --flavor m1.tiny --im
age cirros-0.6.3-x86_64-disk --network dev-net Test1
+-------------------------------------+--------------------------------------+
| Field                               | Value                                |
+-------------------------------------+--------------------------------------+
| OS-DCF:diskConfig                   | MANUAL                               |
| OS-EXT-AZ:availability_zone         | None                                 |
| OS-EXT-SRV-ATTR:host                | None                                 |
| OS-EXT-SRV-ATTR:hostname            | test1                                |
| OS-EXT-SRV-                         | None                                 |
| ATTR:hypervisor_hostname            |                                      |
| OS-EXT-SRV-ATTR:instance_name       | None                                 |
| OS-EXT-SRV-ATTR:kernel_id           | None                                 |
| OS-EXT-SRV-ATTR:launch_index        | None                                 |
| OS-EXT-SRV-ATTR:ramdisk_id          | None                                 |
| OS-EXT-SRV-ATTR:reservation_id      | r-pze8qryd                           |
| OS-EXT-SRV-                         | None                                 |
| ATTR:root_device_name               |                                      |
| OS-EXT-SRV-ATTR:user_data           | None                                 |
| OS-EXT-STS:power_state              | N/A                                  |
| OS-EXT-STS:task_state               | scheduling                           |
```

Repeat creation up to 10 times, changing server name each time.

Step 6: View the instances you've created to verify they were created. Notice how the instances deploy successfully without restriction (up to 10 instances), confirming there is no enforcement boundary on the current configuration.

```
ubuntu@ubuntu:~/Desktop$ openstack server list
+--------------+---------+---------+--------------+--------------+---------+
| ID           | Name    | Status  | Networks     | Image        | Flavor  |
+--------------+---------+---------+--------------+--------------+---------+
| 4a42fea9-    | Test3   | ACTIVE  | dev-net=10   | cirros-      | m1.tiny |
| 3932-4031-   |         |         | .0.20.125    | 0.6.3-       |         |
| 8cc8-        |         |         |              | x86_64-      |         |
| 88fa0d1f4d   |         |         |              | disk         |         |
| 87           |         |         |              |              |         |
| 5254358f-    | Test2   | ACTIVE  | dev-net=10   | cirros-      | m1.tiny |
| 35e2-4a22-   |         |         | .0.20.186    | 0.6.3-       |         |
| bf42-        |         |         |              | x86_64-      |         |
| eb9a348500   |         |         |              | disk         |         |
| 78           |         |         |              |              |         |
| 44349d25-    | Test1   | ACTIVE  | dev-net=10   | cirros-      | m1.tiny |
| 2a23-4e69-   |         |         | .0.20.113    | 0.6.3-       |         |
| b5ce-        |         |         |              | x86_64-      |         |
| 4cce3cec00   |         |         |              | disk         |         |
| 54           |         |         |              |              |         |
| e307b644-    | DEV-APP | SHUTOFF | dev-net=10   | N/A          | m1.tiny |
| e163-40c6-   |         |         | .0.20.23     | (booted      |         |
| b90e-        |         |         |              | from         |         |
| c583e5ff10   |         |         |              | volume)      |         |
| 55           |         |         |              |              |         |
| f45bc2de-    | DEV-DB  | SHUTOFF | dev-net=10   | N/A          | m1.tiny |
| 808d-46ab-   |         |         | .0.20.126,   | (booted      |         |
| bce3-        |         |         | 172.24.4.7   | from         |         |
| 294d37c939   |         |         | 8            | volume)      |         |
| bb           |         |         |              |              |         |
+--------------+---------+---------+--------------+--------------+---------+
```

Notice how the instances deploy successfully without restriction (up to 10 instances).

Step 7: Delete the extra instances you've created. Cleanup returns the environment to the known state from Report #2. The only instances remaining should be the two instances from the previous labs.

```
ubuntu@ubuntu:~/Desktop$ openstack server delete Test1
ubuntu@ubuntu:~/Desktop$ openstack server delete Test2
ubuntu@ubuntu:~/Desktop$ openstack server delete Test3
ubuntu@ubuntu:~/Desktop$ openstack server list
+------------+---------+---------+------------+------------+---------+
| ID         | Name    | Status  | Networks   | Image      | Flavor  |
+------------+---------+---------+------------+------------+---------+
| e307b644-  | DEV-APP | SHUTOFF | dev-net=10 | N/A        | m1.tiny |
| e163-40c6- |         |         | .0.20.23   | (booted    |         |
| b90e-      |         |         |            | from       |         |
| c583e5ff10 |         |         |            | volume)    |         |
| 55         |         |         |            |            |         |
| f45bc2de-  | DEV-DB  | SHUTOFF | dev-net=10 | N/A        | m1.tiny |
| 808d-46ab- |         |         | .0.20.126, | (booted    |         |
| bce3-      |         |         | 172.24.4.7 | from       |         |
| 294d37c939 |         |         | 8          | volume)    |         |
| bb         |         |         |            |            |         |
+------------+---------+---------+------------+------------+---------+
```

The only instances left should be the two instances from the previous labs.

Section 1 Outcome

At this stage:

- Tenant projects can provision resources freely
- No safeguards prevent infrastructure exhaustion
- Governance controls are absent [1]

This establishes the operational baseline prior to quota enforcement.

Section 2: Implementing Project-Level Governance Controls

Purpose of This Section

This section introduces quota limits that align tenant resource consumption with organizational capacity planning and least-privilege infrastructure usage [1].

2.1 Configure Compute Quotas

Step 1: Return to administrator scope. Administrative credentials are required to modify project quotas. This step re-establishes the correct context before applying governance changes.

```
ubuntu@ubuntu:~/Desktop$ source admin-openrc.sh
Please enter your OpenStack Password for project admin as user admin:
ubuntu@ubuntu:~/Desktop$
```

Step 2: Apply compute quotas for the DEV project. This sets a hard ceiling on DEV's compute resource consumption, preventing accidental or deliberate exhaustion of shared compute capacity.

```
ubuntu@ubuntu:~/Desktop$ openstack quota set dev --instances 3 --cores 6 --ram 8192
ubuntu@ubuntu:~/Desktop$
```

What this does:

- Limits DEV to 3 instances maximum
- Restricts to 6 vCPU cores total
- Caps RAM at 8GB (8192 MB)

Step 3: Apply compute quotas for the QA project. QA receives a lower allocation than DEV, reflecting the organization's decision that test environments require fewer resources than active development environments.

```
ubuntu@ubuntu:~/Desktop$ openstack quota set qa --instances 2 --cores 4 --ram 4096
ubuntu@ubuntu:~/Desktop$
```

What this does:

- Limits QA to 2 instances maximum
- Restricts to 4 vCPU cores total
- Caps RAM at 4GB (4096 MB)

2.2 Configure Block Storage Quotas

Step 4: Limit volume usage for DEV. Block storage quotas prevent a single project from consuming the entire Cinder storage pool through excessive volume creation.

```
ubuntu@ubuntu:~/Desktop$ openstack quota set dev --volumes 3 --gigabytes 10
ubuntu@ubuntu:~/Desktop$
```

What this does:

- Allows maximum 3 Cinder volumes
- Total storage limited to 10GB across all volumes

Step 5: Limit volume usage for QA. QA receives a reduced storage allocation consistent with its smaller compute footprint.

```
ubuntu@ubuntu:~/Desktop$ openstack quota set qa --volumes 2 --gigabytes 5
ubuntu@ubuntu:~/Desktop$
```

What this does:

- Allows maximum 2 Cinder volumes
- Total storage limited to 5GB

2.3 Configure Network Resource Quotas

Step 6: Restrict floating IP allocation for both projects. Limiting floating IPs prevents uncontrolled external exposure and conserves the shared pool of externally routable addresses.

```
ubuntu@ubuntu:~/Desktop$ openstack quota set dev --floating-ips 2
ubuntu@ubuntu:~/Desktop$
```

```
ubuntu@ubuntu:~/Desktop$ openstack quota set qa --floating-ips 2
ubuntu@ubuntu:~/Desktop$
```

What this does:

- DEV can allocate 2 floating IPs maximum

- QA can allocate 2 floating IP maximum

Step 7: Limit security group rules for both projects. Limiting the number of security group rules helps keep networks more secure and prevents users from creating too many or overly open rules.

```
ubuntu@ubuntu:~/Desktop$ openstack quota set dev --secgroup-rules 20
ubuntu@ubuntu:~/Desktop$
```

```
ubuntu@ubuntu:~/Desktop$ openstack quota set qa --secgroup-rules 10
ubuntu@ubuntu:~/Desktop$
```

What this does:

- DEV limited to 20 total security group rules
- QA limited to 10 total security group rules

2.4 Verify Applied Quotas

Step 8: Confirm the DEV quotas are properly configured:

```
ubuntu@ubuntu:~/Desktop$ openstack quota show dev
+-----------------------+-------+
| Resource              | Limit |
+-----------------------+-------+
| cores                 |     6 |
| instances             |     3 |
| ram                   |  8192 |
| fixed_ips             |  None |
| networks              |   100 |
| volumes               |     3 |
| snapshots             |    10 |
| gigabytes             |    10 |
| backups               |    10 |
| volumes_lvmdriver-1   |    -1 |
| gigabytes_lvmdriver-1 |    -1 |
| snapshots_lvmdriver-1 |    -1 |
| volumes___DEFAULT__   |    -1 |
| gigabytes___DEFAULT__ |    -1 |
| snapshots___DEFAULT__ |    -1 |
| groups                |    10 |
| check_limit           |  None |
| health_monitors       |  None |
| listeners             |  None |
| load_balancers        |  None |
| l7_policies           |  None |
| pools                 |  None |
| ports                 |   500 |
| project_id            |  None |
| rbac_policies         |    10 |
| routers               |    10 |
| subnets               |   100 |
| subnet_pools          |    -1 |
```

Step 9: Confirm the QA quotas are properly configured:

```
ubuntu@ubuntu:~/Desktop$ openstack quota show qa
+-----------------------+-------+
| Resource              | Limit |
+-----------------------+-------+
| cores                 |     4 |
| instances             |     2 |
| ram                   |  4096 |
| fixed_ips             |  None |
| networks              |   100 |
| volumes               |     2 |
| snapshots             |    10 |
| gigabytes             |     5 |
| backups               |    10 |
| volumes_lvmdriver-1   |    -1 |
| gigabytes_lvmdriver-1 |    -1 |
| snapshots_lvmdriver-1 |    -1 |
| volumes___DEFAULT__   |    -1 |
| gigabytes___DEFAULT__ |    -1 |
| snapshots___DEFAULT__ |    -1 |
| groups                |    10 |
| check_limit           |  None |
| health_monitors       |  None |
| listeners             |  None |
| load_balancers        |  None |
| l7_policies           |  None |
| pools                 |  None |
| ports                 |   500 |
| project_id            |  None |
| rbac_policies         |    10 |
| routers               |    10 |
| subnets               |   100 |
| subnet_pools          |    -1 |
```

Expected output should show your configured limits, including:

- Instances, cores, RAM (compute)
- Volumes, gigabytes (storage)
- Floating IPs, security group rules (network)

Section 2 Outcome

At completion, you'll have:

- Governance controls restricting tenant resource allocation [1]
- Infrastructure capacity is protected from exhaustion [1]
- Project limits aligned with operational roles (DEV gets more resources than QA)

Section 3: Validation and Enforcement Testing

Purpose of This Section

This section validates that OpenStack actively enforces quota restrictions by denying requests exceeding configured limits [1].

3.1 Instance Limit Enforcement

Goal: Verify that DEV cannot create more than 3 instances (the quota you set).

Step 1: Authenticate as DEV user. Switching to a DEV tenant credential confirms that quota enforcement applies to tenant-level operations, not just administrative ones.

```
ubuntu@ubuntu:~/Desktop$ source dev-openrc.sh
Please enter your OpenStack Password for project dev as user DevUser1:
ubuntu@ubuntu:~/Desktop$
```

Step 2: Attempt to create a third instance (should succeed). This confirms that operations within the quota limit are permitted and that the quota was not set too restrictively.

```
ubuntu@ubuntu:~/Desktop$ openstack server create --flavor m1.tiny --image cirros-0.6.3-x86_64-disk --network dev-net quo
ta-test-4
+-------------------------------------+----------------------------------------------------------------------------+
| Field                               | Value                                                                      |
+-------------------------------------+----------------------------------------------------------------------------+
| OS-DCF:diskConfig                   | MANUAL                                                                     |
| OS-EXT-AZ:availability_zone         | None                                                                       |
| OS-EXT-SRV-ATTR:host                | None                                                                       |
| OS-EXT-SRV-ATTR:hostname            | quota-test-4                                                               |
| OS-EXT-SRV-ATTR:hypervisor_hostname | None                                                                       |
| OS-EXT-SRV-ATTR:instance_name       | None                                                                       |
| OS-EXT-SRV-ATTR:kernel_id           | None                                                                       |
| OS-EXT-SRV-ATTR:launch_index        | None                                                                       |
| OS-EXT-SRV-ATTR:ramdisk_id          | None                                                                       |
| OS-EXT-SRV-ATTR:reservation_id      | r-tv9c98hn                                                                 |
| OS-EXT-SRV-ATTR:root_device_name    | None                                                                       |
```

Step 3: Attempt to create a fourth instance (should fail). A quota exceeded error here confirms that OpenStack is actively enforcing the instance limit and that the limit is set correctly.

```
ubuntu@ubuntu:~/Desktop$ openstack server create --flavor m1.tiny --image cirros-0.6.3-x86_64-disk --network dev-net quo
ta-test-4a
ForbiddenException: 403: Client Error for url: http://127.0.0.1/compute/v2.1/servers, Quota exceeded for instances: Requ
ested 1, but already used 3 of 3 instances
ubuntu@ubuntu:~/Desktop$
```

Step 4: Clean up the test instance. Removing the extra instance returns DEV to its expected state and avoids consuming quota needed for subsequent tests.

```
ubuntu@ubuntu:~/Desktop$ openstack server delete quota-test-4
ubuntu@ubuntu:~/Desktop$
```

3.2 Floating IP Enforcement

Goal: Verify floating IP quota limits work.

Step 5: Check current floating IP allocation. Knowing the current count relative to the 2-IP limit determines whether the next allocation attempt will succeed or trigger enforcement.

```
ubuntu@ubuntu:~/Desktop$ openstack floating ip list
+--------------------+---------------------+------------------+--------------------+--------------------+------------------+
| ID                 | Floating IP Address | Fixed IP Address | Port               | Floating Network   | Project          |
+--------------------+---------------------+------------------+--------------------+--------------------+------------------+
| 7e77f5ba-d9a8-     | 172.24.4.78         | 10.0.20.126      | e8dc0430-d24f-     | 24353f0c-c770-     | 4f97f0eebec74d8b8
| 474c-a11d-         |                     |                  | 43ac-a2cb-         | 4e7a-abd7-         | 3b9296bc70dd0bb
| 88239e2e995e       |                     |                  | 9f1dcd5e549e       | 196dcb3bc476       |
+--------------------+---------------------+------------------+--------------------+--------------------+------------------+
ubuntu@ubuntu:~/Desktop$
```

Step 6: Attempt to create another floating IP (this may succeed if currently at 1/2 floating IPs). This tests enforcement behavior by approaching and crossing the configured floating IP limit.

```
ubuntu@ubuntu:~/Desktop$ openstack floating ip create public
Error while executing command: ConflictException: 409, Quota exceeded for resources: ['floatingip'].
ubuntu@ubuntu:~/Desktop$
```

Step 7: Clean up any extra floating IPs created during testing. Releasing unneeded floating IPs returns the project within its allocation and prevents resource waste.

```
ubuntu@ubuntu:~/Desktop$ openstack floating ip delete 172.24.4.137
ubuntu@ubuntu:~/Desktop$
```

3.3 Volume Allocation Enforcement

Goal: Verify storage quota enforcement.

Step 8: Check existing volumes.

```
ubuntu@ubuntu:~/Desktop$ openstack volume list
+--------------------------------------+-------+--------+------+--------------------------------+
| ID                                   | Name  | Status | Size | Attached to                    |
+--------------------------------------+-------+--------+------+--------------------------------+
| 78e7f7e8-8a76-4abd-bcdf-e436d6f1bcd8 | test3 | error  |    1 |                                |
| fade0492-244e-4920-9cc9-af0e059e088f |       | error  |    1 |                                |
| 93d1a3d7-c64c-4d4d-94d2-8bf2abed1016 |       | in-use |    1 | Attached to DEV-APP on /dev/vda |
| cf54859a-3c02-40ab-b02a-80c8894eefab |       | in-use |    1 | Attached to DEV-DB on /dev/vda  |
| 51588218-9389-46e4-9f17-d824588169e2 | Test  | error  |    1 |                                |
+--------------------------------------+-------+--------+------+--------------------------------+
ubuntu@ubuntu:~/Desktop$
```

You should have 2 volumes from previous labs (dev-volume-1 and dev-volume-2), totaling some amount of storage.

Step 9: Attempt to create a volume that would exceed the 10GB quota. A quota exceeded error confirms that Cinder enforces the storage limit and will deny allocations that would push the project over its configured ceiling.

```
ubuntu@ubuntu:~/Desktop$ openstack volume create --size 5 quota-volume-test
HttpException: 413: Client Error for url: http://127.0.0.1/volume/v3/volumes, VolumeLimitExceeded: Maximum number of vol
umes allowed (3) exceeded for quota 'volumes'.
ubuntu@ubuntu:~/Desktop$
```

3.4 Governance Verification

Step 10: Switch to QA project and test their limits. Testing QA separately confirms that each project's quota is enforced independently and that DEV limits do not bleed over into the QA tenant.

```
ubuntu@ubuntu:~/Desktop$ source qa-openrc.sh
Please enter your OpenStack Password for project qa as user QAUser1:
ubuntu@ubuntu:~/Desktop$
```

Step 11: Test QA instance limit (should be 2). Attempting to create instances up to the QA limit validates that the lower QA quota is active and correctly configured.

```
ubuntu@ubuntu:~/Desktop$ openstack server list
+--------------------------+----------+---------+--------------------------+-------------------------+---------+
| ID                       | Name     | Status  | Networks                 | Image                   | Flavor  |
+--------------------------+----------+---------+--------------------------+-------------------------+---------+
| ec9d8b0f-b1b0-4166-a0e7- | TEST-DB  | SHUTOFF | test-net=10.0.10.195,    | N/A (booted from volume) | m1.tiny |
| 60688f7bb11c             |          |         | 172.24.4.107             |                         |         |
| 0925a67e-0b77-4536-8789- | TEST-APP | SHUTOFF | test-net=10.0.10.184,    | N/A (booted from volume) | m1.tiny |
| 5f8207605940             |          |         | 172.24.4.18              |                         |         |
+--------------------------+----------+---------+--------------------------+-------------------------+---------+
ubuntu@ubuntu:~/Desktop$
```

Step 12: If you have 2 instances already, attempt a third. A denied third instance confirms QA cannot allocate excessive resources.

```
ubuntu@ubuntu:~/Desktop$ openstack server create --flavor m1.tiny --image cirros-0.6.3-x86_64-disk --network qa-net quot
a-test-3
ForbiddenException: 403: Client Error for url: http://127.0.0.1/image/v2/images?name=cirros-0.6.3-x86_64-disk, You are n
ot authorized to complete get_images action.<br /><br />

ubuntu@ubuntu:~/Desktop$
```

QA cannot allocate excessive networking resources

Section 3 Outcome

- **Quota enforcement is active** - requests exceeding limits are denied
- **Different projects have different limits** - DEV gets more resources than QA
- **Infrastructure is protected** - no single tenant can exhaust shared resources
- **OpenStack governance works as designed** - consistent with enterprise practices

Validation confirms that OpenStack governance controls successfully prevent tenant resource exhaustion while allowing authorized workloads to function normally [1] [2] [3].

Quota enforcement provides operational stability consistent with enterprise cloud deployment practices [1].

Section 4: Operational Governance Monitoring and Capacity Validation

Purpose of This Section

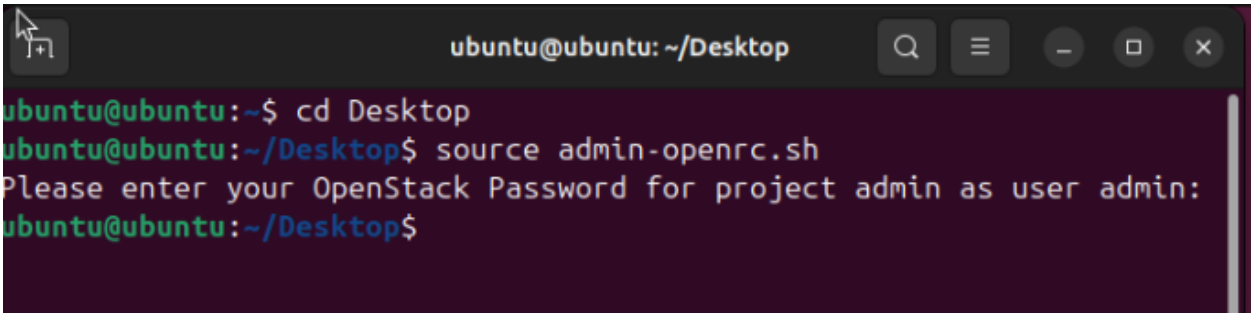
While previous sections verified that quota limits deny excessive resource allocation requests, enterprise cloud environments must also provide administrators with visibility into tenant resource consumption. Governance controls are most effective when administrators can monitor usage trends, identify approaching limits, and validate that infrastructure capacity remains available for authorized workloads.

This section demonstrates how OpenStack administrative tools can be used to monitor quota utilization, verify remaining capacity, and validate that governance controls actively preserve infrastructure availability across tenant projects [1].

4.1 Review Project Resource Utilization

Goal: Verify current resource consumption relative to configured quota limits.

Step 1: Authenticate as administrator. Administrative scope is required to view cross-project quota utilization data.



Step 2: Display quota usage for the DEV project. This surfaces current in-use values alongside configured limits, allowing administrators to identify projects approaching their ceilings before an exhaustion event occurs.

```
ubuntu@ubuntu:~/Desktop$ openstack quota show --usage dev
+-----------------------+-------+--------+----------+
| Resource              | Limit | In Use | Reserved |
+-----------------------+-------+--------+----------+
| cores                 |     6 |      2 |        0 |
| instances             |     3 |      2 |        0 |
| ram                   |  8192 |   1024 |        0 |
| fixed_ips             |     0 |      0 |        0 |
| networks              |   100 |      1 |        0 |
| volumes               |     3 |      5 |        0 |
| snapshots             |    10 |      0 |        0 |
| gigabytes             |    10 |      5 |        0 |
| backups               |    10 |      0 |        0 |
| volumes_lvmdriver-1   |    -1 |      4 |        0 |
| gigabytes_lvmdriver-1 |    -1 |      4 |        0 |
| snapshots_lvmdriver-1 |    -1 |      0 |        0 |
| volumes___DEFAULT__   |    -1 |      1 |        0 |
| gigabytes___DEFAULT__ |    -1 |      1 |        0 |
| snapshots___DEFAULT__ |    -1 |      0 |        0 |
| groups                |    10 |      0 |        0 |
| ports                 |   500 |      4 |        0 |
| rbac_policies         |    10 |      0 |        0 |
| routers               |    10 |      1 |        0 |
| subnets               |   100 |      1 |        0 |
| subnet_pools          |    -1 |      0 |        0 |
| injected-file-size    | 10240 |      0 |        0 |
| injected-path-size    |   255 |      0 |        0 |
| injected-files        |     5 |      0 |        0 |
| key-pairs             |   100 |      0 |        0 |
| properties            |   128 |      0 |        0 |
| server-group-members  |    10 |      0 |        0 |
| server-groups         |    10 |      0 |        0 |
| floating-ips          |     2 |      0 |        0 |
| secgroup-rules        |    20 |      0 |        0 |
| secgroups             |    10 |      0 |        0 |
| backup-gigabytes      |  1000 |      0 |        0 |
| per-volume-gigabytes  |    -1 |      0 |        0 |
+-----------------------+-------+--------+----------+
```

Step 3: Repeat for the QA project. Comparing utilization across both projects gives a full picture of infrastructure consumption and remaining capacity.

```
ubuntu@ubuntu:~/Desktop$ openstack quota show --usage qa
+-----------------------+-------+--------+----------+
| Resource              | Limit | In Use | Reserved |
+-----------------------+-------+--------+----------+
| cores                 |     4 |      2 |        0 |
| instances             |     2 |      2 |        0 |
| ram                   |  4096 |   1024 |        0 |
| fixed_ips             |     0 |      0 |        0 |
| networks              |   100 |      1 |        0 |
| volumes               |     2 |      3 |        0 |
| snapshots             |    10 |      0 |        0 |
| gigabytes             |     5 |      3 |        0 |
| backups               |    10 |      0 |        0 |
| volumes_lvmdriver-1   |    -1 |      3 |        0 |
| gigabytes_lvmdriver-1 |    -1 |      3 |        0 |
| snapshots_lvmdriver-1 |    -1 |      0 |        0 |
| volumes___DEFAULT__   |    -1 |      0 |        0 |
| gigabytes___DEFAULT__ |    -1 |      0 |        0 |
| snapshots___DEFAULT__ |    -1 |      0 |        0 |
| groups                |    10 |      0 |        0 |
| ports                 |   500 |      4 |        0 |
| rbac_policies         |    10 |      0 |        0 |
| routers               |    10 |      1 |        0 |
| subnets               |   100 |      1 |        0 |
| subnet_pools          |    -1 |      0 |        0 |
| injected-file-size    | 10240 |      0 |        0 |
| injected-path-size    |   255 |      0 |        0 |
| injected-files        |     5 |      0 |        0 |
| key-pairs             |   100 |      0 |        0 |
| properties            |   128 |      0 |        0 |
| server-group-members  |    10 |      0 |        0 |
| server-groups         |    10 |      0 |        0 |
| floating-ips          |     2 |      0 |        0 |
| secgroup-rules        |    10 |      0 |        0 |
| secgroups             |    10 |      0 |        0 |
| backup-gigabytes      |  1000 |      0 |        0 |
| per-volume-gigabytes  |    -1 |      0 |        0 |
+-----------------------+-------+--------+----------+
```

Expected Observation

The output displays:

- Resource limits
- Resources currently in use
- Remaining available capacity

OpenStack provides quota usage visibility to assist administrators in maintaining operational stability and preventing tenant resource exhaustion [1].

4.2 Validate Remaining Compute Capacity

Goal: Confirm governance controls preserve infrastructure resources for future workloads.

Step 4: Switch to DEV project credentials. Scoping to DEV confirms that the following actions are performed within the tenant's quota context.

```
ubuntu@ubuntu:~/Desktop$ source dev-openrc.sh
Please enter your OpenStack Password for project dev as user DevUser1:
```

Step 5: List currently deployed instances. Comparing the current instance count against the configured limit confirms how much quota capacity remains before the ceiling is reached.

```
ubuntu@ubuntu:~/Desktop$ openstack server list
+--------------------------------------+---------+---------+--------------------------------+-------------------------+---------+
| ID                                   | Name    | Status  | Networks                       | Image                   | Flavor  |
+--------------------------------------+---------+---------+--------------------------------+-------------------------+---------+
| e307b644-e163-40c6-b90e-c583e5ff1055 | DEV-APP | SHUTOFF | dev-net=10.0.20.23             | N/A (booted from volume) | m1.tiny |
| f45bc2de-808d-46ab-bce3-294d37c939bb | DEV-DB  | SHUTOFF | dev-net=10.0.20.126, 172.24.4.78 | N/A (booted from volume) | m1.tiny |
+--------------------------------------+---------+---------+--------------------------------+-------------------------+---------+
```

Step 6: Attempt deployment within remaining quota allowance.

```
ubuntu@ubuntu:~/Desktop$ openstack server create --flavor m1.tiny --image cirros-0.6.3-x86_64-disk --network dev-net governance-test-instance
+-------------------------------------+-----------------------------------------
```

```
ubuntu@ubuntu:~/Desktop$ openstack quota show --usage dev
+-----------------------+-------+--------+----------+
| Resource              | Limit | In Use | Reserved |
+-----------------------+-------+--------+----------+
| cores                 |     6 |      3 |        0 |
| instances             |     3 |      3 |        0 |
| ram                   |  8192 |   1536 |        0 |
| fixed_ips             |     0 |      0 |        0 |
| networks              |   100 |      1 |        0 |
| volumes               |     3 |      5 |        0 |
| snapshots             |    10 |      0 |        0 |
| gigabytes             |    10 |      5 |        0 |
| backups               |    10 |      0 |        0 |
| volumes_lvmdriver-1   |    -1 |      4 |        0 |
| gigabytes_lvmdriver-1 |    -1 |      4 |        0 |
| snapshots_lvmdriver-1 |    -1 |      0 |        0 |
| volumes___DEFAULT__   |    -1 |      1 |        0 |
| gigabytes___DEFAULT__ |    -1 |      1 |        0 |
| snapshots___DEFAULT__ |    -1 |      0 |        0 |
| groups                |    10 |      0 |        0 |
| ports                 |   500 |      5 |        0 |
| rbac_policies         |    10 |      0 |        0 |
| routers               |    10 |      1 |        0 |
| subnets               |   100 |      1 |        0 |
| subnet_pools          |    -1 |      0 |        0 |
| injected-file-size    | 10240 |      0 |        0 |
| injected-path-size    |   255 |      0 |        0 |
| injected-files        |     5 |      0 |        0 |
| key-pairs             |   100 |      0 |        0 |
| properties            |   128 |      0 |        0 |
| server-group-members  |    10 |      0 |        0 |
| server-groups         |    10 |      0 |        0 |
| floating-ips          |     2 |      0 |        0 |
| secgroup-rules        |    20 |      0 |        0 |
| secgroups             |    10 |      0 |        0 |
| backup-gigabytes      |  1000 |      0 |        0 |
| per-volume-gigabytes  |    -1 |      0 |        0 |
+-----------------------+-------+--------+----------+
```

Quota tracking ensures infrastructure availability is preserved while permitting authorized workloads to operate normally [1].

4.3 Monitor Network Resource Consumption

Goal: Validate governance visibility for externally exposed networking resources.

Step 9: Display floating IP utilization. Monitoring floating IP consumption identifies projects that are approaching the external exposure limit, which is a security and availability concern.

```
ubuntu@ubuntu:~/Desktop$ openstack subnet list --project dev
+--------------------------------------+------------+--------------------------------------+--------------+
| ID                                   | Name       | Network                              | Subnet       |
+--------------------------------------+------------+--------------------------------------+--------------+
| 9a8c2511-facf-4d12-bdaf-cb5a33ef1bec | dev-subnet | a729b745-9ead-4256-b0b5-1ce16ef6d2f7 | 10.0.20.0/24 |
+--------------------------------------+------------+--------------------------------------+--------------+
```

Step 10: Check quota usage again.

```
ubuntu@ubuntu:~/Desktop$ openstack quota show --usage dev
+-----------------------+-------+--------+----------+
| Resource              | Limit | In Use | Reserved |
+-----------------------+-------+--------+----------+
| cores                 |     6 |      3 |        0 |
| instances             |     3 |      3 |        0 |
| ram                   |  8192 |   1536 |        0 |
| fixed_ips             |     0 |      0 |        0 |
| networks              |   100 |      1 |        0 |
| volumes               |     3 |      5 |        0 |
| snapshots             |    10 |      0 |        0 |
| gigabytes             |    10 |      5 |        0 |
| backups               |    10 |      0 |        0 |
| volumes_lvmdriver-1   |    -1 |      4 |        0 |
| gigabytes_lvmdriver-1 |    -1 |      4 |        0 |
| snapshots_lvmdriver-1 |    -1 |      0 |        0 |
| volumes___DEFAULT__   |    -1 |      1 |        0 |
| gigabytes___DEFAULT__ |    -1 |      1 |        0 |
| snapshots___DEFAULT__ |    -1 |      0 |        0 |
| groups                |    10 |      0 |        0 |
| ports                 |   500 |      5 |        0 |
| rbac_policies         |    10 |      0 |        0 |
| routers               |    10 |      1 |        0 |
| subnets               |   100 |      1 |        0 |
```

Observe subnet allocation relative to configured limits.

Networking quotas restrict excessive external exposure and maintain controlled access to shared network infrastructure [3].

4.4 Validate Storage Capacity Governance

Goal: Confirm block storage usage remains constrained by quota enforcement.

Step 11: Display existing volumes. Listing current volumes with their sizes establishes the total allocated storage against which the quota ceiling can be checked.

```
ubuntu@ubuntu:~/Desktop$ openstack volume list
+--------------------------------------+---------------+-----------+------+-------------+
| ID                                   | Name          | Status    | Size | Attached to |
+--------------------------------------+---------------+-----------+------+-------------+
| ef5ba569-b679-4d23-8470-3b12ed119673 | test-443      | available |    2 |             |
| 6c6b941f-36cf-4a38-8080-770abd76e772 | test-44       | error     |    2 |             |
| 099b6c80-de28-49c1-92d0-fafee652a24d | test-4        | error     |    2 |             |
| 26e49a1f-f36f-4f25-8275-56590d06bb75 | test-volume-2 | available |    1 |             |
| 3f0d1364-d5ec-44bd-9e11-7578a5d06495 | test-volume   | error     |    1 |             |
+--------------------------------------+---------------+-----------+------+-------------+
```

Step 12: Check storage quota utilization.

```
ubuntu@ubuntu:~/Desktop$ openstack quota show --usage dev
+----------------------+-------+--------+----------+
| Resource             | Limit | In Use | Reserved |
+----------------------+-------+--------+----------+
| cores                |     6 |      3 |        0 |
| instances            |     3 |      3 |        0 |
| ram                  |  8192 |   1536 |        0 |
| fixed_ips            |     0 |      0 |        0 |
| networks             |   100 |      1 |        0 |
| volumes              |     3 |      5 |        0 |
| snapshots            |    10 |      0 |        0 |
| gigabytes            |    10 |      5 |        0 |
| backups              |    10 |      0 |        0 |
```

Document:

- Total allocated gigabytes
- Remaining storage capacity

Block storage quota enforcement prevents uncontrolled volume allocation and protects shared storage availability within OpenStack environments [2].

Section 4 Outcome

At completion:

- Administrators can monitor tenant resource consumption in real time
- Remaining infrastructure capacity is visible and measurable
- Governance controls actively preserve compute, network, and storage availability
- Operational visibility supports proactive cloud administration and capacity planning [1], [2], [3]

Engagement Summary

Through this engagement, Southriver Software Solutions transitioned from an unrestricted development cloud toward a governed multi-tenant infrastructure model. Resource quotas now prevent accidental denial-of-service conditions caused by uncontrolled provisioning while preserving tenant autonomy [1] [3].

This governance layer complements previously implemented identity and network security controls and prepares the environment for upcoming engagements involving encryption, monitoring, and incident response readiness.

References

[1] OpenStack Foundation, “Manage Compute Service Quotas,” OpenStack Nova Administration Guide, 2024. [Online]. Available: https://docs.openstack.org/nova/latest/admin/quotas.html

[2] OpenStack Foundation, “Block Storage Service Quotas,” OpenStack Cinder Administration Guide, 2024. [Online]. Available: https://docs.openstack.org/cinder/latest/admin/quotas.html

[3] OpenStack Foundation, “Quota Management for Networking,” OpenStack Neutron Administration Guide, 2024. [Online]. Available: https://docs.openstack.org/neutron/latest/admin/ops-quotas.html